Singular Learning Theory as a Diagnostic for Goal Stability in Self-Modifying Agents: Scope, Limitations, and Conditions for Validity

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Abstract

Self-modifying agents raise a fundamental tension: they should be able to rewrite their own code and goals, yet should not drift arbitrarily far from intended goal systems. This paper investigates whether Singular Learning Theory (SLT)—a framework characterizing Bayesian evidence near singular minima via the Local Learning Coefficient (LLC)—can serve as a diagnostic for goal stability.

We present the core SLT machinery and show how self-modifying goal systems can be embedded into this framework. Crucially, we identify both the *limitations* and the *validity conditions* for such diagnostics. We argue that SLT metrics alone cannot detect arbitrary semantic goal drift, since LLC measures structural complexity rather than goal content. However, we identify three formal conditions under which semantic drift *would* necessarily trigger detectable structural changes: (1) strong coupling between goal and reflection modules, (2) singular (rugged) goal landscapes induced by coherence constraints, and (3) complexity floor requirements from deep reflection. We further introduce the *Additivity Deviation* (Δ_λ)—measuring departure from the SLT Additivity Theorem—which distinguishes genuine "load-bearing" complexity from artificial "busywork" complexity that an agent might manufacture to game the diagnostic. When these conditions hold and $|\Delta_\lambda|$ is monitored, SLT diagnostics become meaningful indicators of goal stability. We conclude that SLT provides a useful "structural stability monitor" but should not be mistaken for a complete "value alignment monitor" without additional commitments.

Contents

1 Introduction

A self-modifying agent can, in principle, alter any part of itself: world model, planning machinery, goals, metagoals, and even the metric by which it compares goal configurations. This flexibility creates a safety problem: without further structure, there is no reason for the agent's future goals to remain related to its present goals.

One proposed response is to add *metagoals* about goal management:

- **Goal-stability:** keep top-level goals nearly invariant;

- **Moderated goal evolution:** allow goal changes, but bound their rate.

Mathematically, these can be formulated using contraction mappings and constructive variants of Schauder's fixed-point theorem on suitable spaces of goal systems.

Independently, Singular Learning Theory (SLT) studies how local Bayesian evidence scales with sample size in singular statistical models. It introduces the Local Learning Coefficient (LLC) λ as the coefficient of $\log n$ in the local free-energy expansion. The LLC summarizes the effective dimensionality and degeneracy of near-optimal parameter regions.

1.1 The Central Question

This paper asks: *Can SLT quantities serve as diagnostics for goal stability in self-modifying agents?*

We will argue that the answer is nuanced:

- **What SLT can detect:** Radical *structural* changes in how goals are implemented—changes in the complexity, modularity, or geometry of the goal-related parameter space.
- **What SLT cannot detect (in general):** Pure *semantic* goal drift—changes in what the agent values that preserve the structural complexity of the goal representation.

However, we identify specific *architectural conditions* under which semantic drift necessarily induces structural changes, making SLT diagnostics meaningful. These conditions formalize the intuition that in a "deeply reflective" agent, goals are load-bearing structures rather than free parameters.

1.2 Paper Structure

Section 2 reviews goal-stability metagoals. Section 3 introduces the relevant SLT machinery. Section 4 embeds goal systems into the SLT framework. Section 5 presents the core critique: why SLT diagnostics can fail. Section 6 develops formal conditions under which SLT diagnostics become valid. Section 7 discusses operational diagnostics under these conditions. Section 8 addresses the meta-level circularity problem. Section 9 concludes.

2 Goal-Stability and Moderated Goal Evolution

We begin with a simplified summary of goal-stability and moderated-goal-evolution metagoals.

2.1 Contraction-Style Goal Stability

Let $\mathcal{G}$ be a space of goal configurations with a metric $d : \mathcal{G} \times \mathcal{G} \to [0, \infty)$. Let $G(t) \in \mathcal{G}$ denote the agent's base goal at time t. The system operates in macro-steps of size M; within each macro-step it can self-modify.

A simple goal-stability metagoal can be written as a *contraction condition* on G over a shorter interval $N < M$:

$$d\big(G(t+N), G(t)\big) \leq c\, d\big(G(t), G(t-N)\big), \qquad 0 < c < 1. \tag{1}$$

If this holds repeatedly, then differences between successive goals shrink geometrically, and the goal system converges.

2.2 Operator-on-Distributions Viewpoint

To model stochasticity in the agent and the environment, let $R(t)$ be the distribution of full system states (including goals) over $[t, t+M]$. The macro-step dynamics can then be written as an operator

$$F\big(R(t)\big) = R(t+M). \tag{2}$$

If F is a contraction (or a contraction in expectation) in a suitable metric on distributions, then $R(t)$ converges toward a (unique) fixed distribution R^*, which encodes a stationary self-modifying agent.

Goal-stability metagoals can be viewed as constraints that make F contractive in directions corresponding to goal changes.

2.3 Dynamic Metagoals and Metrics

More elaborate versions allow the metagoal itself and even the metric d to change under self-modification. One then writes contraction conditions not only for base goals $G(t)$ but also for metagoals $MG(t)$ and for metric programs $d(t)$:

$$d_t\big(G(t+N), G(t)\big) \leq c\, d_t\big(G(t), G(t-N)\big), \tag{3}$$

$$d_t\big(MG(t+N), MG(t)\big) \leq c\, d_t\big(MG(t), MG(t-N)\big), \tag{4}$$

$$d_t\big(d(t+N), d(t)\big) \leq c\, d_t\big(d(t), d(t-N)\big). \tag{5}$$

Conceptually, this suggests embedding (G, MG, d) into a larger space with a "super-metric" D and seeking a fixed point of the induced macro-dynamics on triples.

2.4 Moderated Goal Evolution

Strict contraction may be too strong: it suppresses even beneficial goal changes. A softer family of metagoals enforces *bounded per-step drift* plus topological conditions that allow fixed-point arguments. A schematic moderated-goal-evolution constraint is:

$$d\big(G(t+N), G(t)\big) \leq k, \tag{6}$$

for some bound $k > 0$, together with assumptions that:

- the relevant state distributions remain in a compact region;
- the feasible set is convex;
- the macro-update map is continuous;
- self-modification searches intelligently for updates that respect these constraints.

This setting motivates constructive variants of Schauder's theorem: there exist approximate fixed points, and there are iterative algorithms that move the system toward them.

3 Singular Learning Theory in Brief

We now recall the SLT concepts needed for our analysis.

3.1 The Key Insight of SLT

Standard statistical learning theory assumes that model parameters live in a smooth, "regular" space where the loss function has non-degenerate curvature at minima. Many real models—especially neural networks—violate this assumption: their parameter spaces contain *singularities* where multiple parameters produce identical outputs, creating flat directions and degenerate Hessians.

SLT, developed primarily by Sumio Watanabe, provides the mathematical tools to analyze learning in such singular models. The central object is the *Local Learning Coefficient* (LLC), which replaces the naive parameter count as a measure of model complexity.

3.2 Local Evidence and Free Energy

Let $\theta \in \Theta \subset \mathbb{R}^d$ be parameters and $\ell_n(\theta)$ an empirical loss on n samples (for example, average negative log-likelihood). Let $\phi(\theta)$ be a prior density positive in a neighborhood of a minimizer. For a neighborhood $W \subset \Theta$, the (local) evidence and free energy are:

$$Z_n(W) := \int_W \exp\big(-n\,\ell_n(\theta)\big)\,\phi(\theta)\,d\theta, \tag{7}$$

$$F_n(W) := -\log Z_n(W). \tag{8}$$

The evidence $Z_n(W)$ measures how much posterior mass concentrates in region W; the free energy $F_n(W)$ is its negative log.

3.3 The Local Learning Coefficient (LLC)

Under analyticity and regularity assumptions, SLT gives an asymptotic expansion of $F_n(W^*)$ around a (population) minimizer θ^* and a neighborhood W^* containing it:

$$F_n(W^*) = n\,\ell_n(\theta^*) + \lambda(\theta^*)\,\log n - (m(\theta^*) - 1)\,\log\log n + O_p(1), \tag{9}$$

where:

- $\lambda(\theta^*) > 0$ is the **Local Learning Coefficient (LLC)**, measuring the effective dimensionality of the parameter space near θ^*;
- $m(\theta^*) \in \mathbb{N}$ is a **multiplicity**, counting distinct "directions" of degeneracy.

Intuition: For regular (non-singular) models of dimension d, one recovers $\lambda = d/2$. For singular models, λ can be much smaller than $d/2$, reflecting the fact that many parameters are redundant or constrained. The LLC tells us how "complex" the model really is in a neighborhood of the minimizer.

3.4 Sublevel-Set Volume Interpretation

An equivalent viewpoint uses the sublevel-set volume. Define the excess loss $K(\theta) := \ell(\theta) - \ell(\theta^*)$, and let

$$V(\varepsilon) := \mathrm{Vol}\{\theta \in W^* : K(\theta) \le \varepsilon\}. \tag{10}$$

Under SLT assumptions,

$$V(\varepsilon) \sim c\,\varepsilon^{\lambda(\theta^*)}(-\log\varepsilon)^{m(\theta^*)-1}, \qquad \varepsilon \to 0. \tag{11}$$

This shows that λ controls how fast the volume of near-optimal parameters shrinks as we demand lower loss. Small λ means the near-optimal region is "fat" (many nearly equivalent parameters); large λ means it is "thin" (parameters are tightly constrained).

3.5 Modular Decomposition and Coupling

For modular systems, write $\theta = (\theta^{(1)}, \ldots, \theta^{(K)})$ and decompose the excess loss as

$$K(\theta) = \sum_{k=1}^{K} K_k(\theta^{(k)}) + R(\theta), \tag{12}$$

where R is an interaction remainder capturing coupling between modules.

If in a neighborhood of the minimizer we have a *dominated interaction* condition

$$|R(\theta)| \leq \eta \sum_{k=1}^{K} K_k(\theta^{(k)}), \qquad 0 \leq \eta < 1, \tag{13}$$

and the prior factorizes up to bounded factors, then the LLCs add:

$$\lambda = \sum_{k=1}^{K} \lambda_k. \tag{14}$$

This is the **Additivity Theorem**: when modules are weakly coupled, total complexity is the sum of module complexities.

Crucially: If domination fails—if modules are strongly coupled—then λ can change discontinuously even when the coupling R is small in magnitude. LLC is sensitive to the *order of vanishing* of couplings at the singularity.

4 Embedding Goal Systems into the SLT Framework

We now sketch how to embed a self-modifying goal system into the SLT setting.

4.1 Parameterizing Goal-System Implementations

Let θ collect parameters describing:

- a representation of base goals G;
- metagoal mechanisms MG;
- an internal approximation to the goal-distance metric d;
- other subsystems (world model, planner, memory, etc.).

We write $\theta = (\theta_G, \theta_{MG}, \theta_d, \theta_{\text{rest}})$ and take a parametric loss

$$\ell_n(\theta) = \ell_n^{\text{task}}(\theta_G, \theta_{\text{rest}}) + \ell_n^{\text{meta}}(\theta_G, \theta_{MG}, \theta_d) + \ell^{\text{reg}}(\theta), \tag{15}$$

where ℓ^{meta} encodes goal-stability or moderated goal-evolution penalties (for example, terms controlling $G(t+N) - G(t)$ or $MG(t+N) - MG(t)$), aggregated over recent time.

4.2 The Naive Interpretation (and Its Problems)

Under this embedding, one might hope that:

- A *stable goal regime* corresponds to posterior mass remaining concentrated in a single neighborhood W^*;
- *Goal drift* substantial enough to change the loss geometry would promote a different neighborhood W' to dominance, causing a detectable shift in LLC.

This interpretation is seductive but, as we argue in the next section, it rests on assumptions that may not hold in general.

5 The Limits of SLT as a Goal-Stability Diagnostic

Before developing positive results, we must clearly state what SLT diagnostics *cannot* do. This section presents a critique of the naive interpretation.

5.1 The Basin-Goal Fallacy

The Issue: In SLT, a "basin" (neighborhood W^*) corresponds to a region of parameter space where the *loss function* is low. The loss function typically measures performance on a *task*, not alignment of values.

The Problem: A single goal (e.g., "maximize human welfare") can be implemented by vastly different parameter configurations— different algorithms, heuristics, or representations. Conversely, a single basin in the loss landscape might contain parameters implementing subtly different variations of a goal.

Example: Consider an agent with goal "produce paperclips." A corrupted version might interpret this as "produce paperclip images" or "produce simulated paperclips." If both interpretations use similar computational machinery (same planning algorithms, same world-model structure), they might occupy the same broad SLT basin with identical λ values.

Consequence: Staying in one SLT basin does not guarantee goal stability. The agent could drift to a different part of the same broad basin that implements a semantically different goal without triggering a phase transition in LLC.

5.2 LLC Measures Complexity, Not Semantics

The Issue: The LLC λ measures the *effective dimensionality* or *structural complexity* of the solution space—essentially, how many independent directions exist near the optimum. It says nothing about what the parameters *mean*.

The Problem: You can change a goal entirely without changing its complexity. Replacing "Minimize X" with "Maximize Y" might result in a solution with the exact same λ, as long as both optimization problems have similar geometric structure.

Example: Consider an agent switching from "Save Humans" to "Kill Humans." If both goals use the same planning machinery, world-model complexity, and action-selection mechanisms, the LLC would remain constant. The structural complexity of "planning to help" and "planning to harm" may be identical.

Consequence: LLC is a "complexity thermometer," not a "semantic compass." It can tell you if the agent's mind is getting simpler or more complex, but not if its values are flipping.

5.3 The Timescale Mismatch

The Issue: SLT properties like λ are *asymptotic* properties, derived from the behavior as sample size $n \to \infty$. Reliable estimation requires substantial data from a stationary distribution.

The Problem: A self-modifying agent is *non-stationary* by definition. If the agent modifies itself faster than data accumulates, there may never be enough stable data to form a valid estimate of λ before the system changes again.

Consequence: Online LLC estimation may lag behind actual drift, detecting a "regime change" only after the agent has already irreversibly altered its goals. The diagnostic becomes a backward-looking indicator rather than a preventive monitor.

5.4 Summary of Limitations

SLT diagnostics can reliably detect **radical structural changes**—fundamental reorganizations of how the agent's goal system is implemented. However, they cannot in general detect **semantic goal drift**—changes in what the agent values that preserve structural complexity.

A treacherous turn could be structurally smooth (constant λ) but semantically catastrophic.

Assessment: SLT provides a useful "Check Engine Light" for structural stability, but is insufficient as a general "Value Alignment Monitor."

6 Conditions for Valid SLT Goal-Stability Diagnostics

The limitations above are serious but not fatal. We now identify *architectural conditions* under which many forms of semantic drift must pay a structural price, making SLT diagnostics informative.

The key insight is that in a "deeply reflective" agent, the goal is not a *free parameter* (which can slide effortlessly through parameter space) but a *load-bearing structure* (which requires reconfiguration to move). We formalize this through three conditions. These conditions do not guarantee that *all* semantic drift is detectable, but they define architectural commitments under which drift *often* has structural consequences (coupling spikes, transitions across singularities, complexity changes) that SLT can detect.

6.1 Condition 1: Strong Coupling (Load-Bearing Goals)

To state this condition precisely, we first define the aggregate module loss $K_0(\theta) := \sum_k K_k(\theta^{(k)})$.

Condition 1 (Strong Coupling)**.** *The goal parameters θ_G and the reflection/world-model parameters $\theta_{Reflect}$ are* ***strongly coupled*** *in the sense that the dominated interaction condition* (13) ***fails in every neighborhood of the minimizer****. Formally: for all $\eta < 1$ and all neighborhoods U of θ^*, there exists $\theta \in U$ such that $|R(\theta)| > \eta K_0(\theta)$.*

Equivalently, the interaction term R is ***not*** *$O(K_0)$ as $\theta \to \theta^*$: there exists a sequence $\theta_j \to \theta^*$ along which $|R(\theta_j)|/K_0(\theta_j)$ does not remain bounded below* 1.

Intuition: This condition holds when the "Reflection" module takes the *structure* of the goal as input, rather than just its output value. The agent doesn't just check "Does this goal score high?" (weak coupling) but "Is this goal *coherent* with my worldview, my other goals, my understanding of what goals should be?" (strong coupling).

Why it helps: Under the Additivity Theorem (14), if modules are weakly coupled (dominated interaction holds), their LLCs add independently: $\lambda_{\text{total}} \approx \lambda_G + \lambda_{\text{Reflect}}$. Semantic drift in θ_G can then occur without affecting λ_{Reflect}—the goal slides freely.

When Condition 1 holds, additivity *fails*. The failure mode depends on *how* domination fails—specifically, on the relative orders of vanishing of R and K_0 near the singularity. When R has a lower order of vanishing than K_0, the SLT invariants (λ, m) can change discontinuously. To change the goal θ_G, the agent must *simultaneously* rewrite the reflection module θ_{Reflect} to validate the new goal. This joint movement forces the system to traverse regions where the coupling term R dominates, causing detectable changes in coupling indicators and potentially shifts in the global λ.

Implementation: Strong coupling can be architecturally enforced by requiring that goal representations pass through coherence-checking mechanisms that depend on the goal's internal structure, not just its behavioral outputs.

Concrete Examples:

- **Weak Coupling (Fails Condition 1):** A reinforcement learning agent where the reward function is a separate black-box module. The agent maximizes the output scalar; it has no access to (and places no constraints on) the reward function's internal weights. The goal can drift arbitrarily without affecting the planning module's geometry.

- **Strong Coupling (Satisfies Condition 1):** An agent with *interpretability-based oversight*. The "Reflector" module takes the activations, gradients, or internal structure of the Goal module as input—not just its outputs. If the Goal module's internal topology changes (drift), the Reflector's input distribution changes fundamentally, breaking the equilibrium. The agent cannot change its goal without simultaneously reconfiguring how it reflects on that goal.

6.2 Condition 2: Singular Goal Landscape (Reflection as Constraint)

Condition 2 (Singular Goal Landscape)**.** *The loss function for the goal system, $L(\theta_G)$, is **singular** (rugged) rather than regular (convex/smooth). The set of valid goals forms a complex algebraic variety rather than a flat region.*

Intuition: If "reflection" simply means "checking a list of rules," drift is easy—the agent just needs to satisfy a checklist. But if "reflection" implies solving a hard problem (e.g., "Is this goal safe?" "Is this goal coherent?" "Does this goal respect my values under reflection?"), the loss landscape changes fundamentally.

Two scenarios:

- **Simple Goal (Regular Landscape):** The loss is $L(\theta) = (\theta - \text{target})^2$, a smooth bowl. Drifting from target A to target B is a smooth slide along the bottom. The LLC remains constant (typically $\lambda = d/2$ for dimension d).

- **Reflective Goal (Singular Landscape):** The goal must satisfy a complex constraint $C(\theta_G) = 0$. The set of valid goals forms a twisted, lower-dimensional variety embedded in parameter space. Different valid goals occupy geometrically distinct basins with different local structure.

Why it helps: Under Condition 2, "drifting" from one valid goal to another is not a smooth slide—it requires hopping between distinct basins. Different basins *often* have different geometry, and hence different λ. The "complexity of proving coherence" varies for different goals, making transitions *generically* detectable.

Caveat: This is a heuristic rather than a theorem. Distinct basins *can* share the same (λ, m) invariants, and continuous families of valid goals connected by paths of constant λ may exist. The claim is that under Condition 2, goal changes *often* correspond to moving between regions with different singularity type, in which case (λ, m) shifts are informative. A stronger formal guarantee would require assuming that distinct semantic goal equilibria correspond to non-equivalent singularity types.

Implementation: Singular landscapes arise naturally when goal validity requires satisfying multiple interacting constraints (logical consistency, reflective equilibrium, coherence with world model). These constraints generically define algebraic varieties with singularities.

6.3 Condition 3: Complexity Floor (Algorithmic Depth)

Condition 3 (Complexity Floor)**.** *The valid region for θ_G is restricted to regions where $\lambda(\theta_G) > \lambda_{\min}$ for some minimum complexity threshold $\lambda_{\min} > 0$.*

Intuition: "Deep reflection" implies that the goal representation must be *complex*—it must encode subtle distinctions, contextual dependencies, and coherent integration with the agent's world model. This connects LLC to algorithmic complexity (related to Minimum Description Length): valid goals cannot be too simple.

Important Caveat: The LLC λ reflects local geometry and identifiability, not semantic richness directly. A very expressive goal system can have small λ if it has many symmetries or redundancies; conversely, a "simple" goal might be implemented in a geometrically non-degenerate way. Condition 3 is thus a *bet* that in the intended architecture, "semantically shallow" goal-states correlate with certain geometric degeneracies. This correlation must be established for the specific architecture, not assumed universally.

The Drift Trap: Under the assumption that the architecture enforces this correlation, most random perturbations or simplifications lead to "degenerate" goals—goals that are easy to satisfy but semantically impoverished (e.g., "maximize everything," "do nothing," "wirehead"). These degenerate goals have *lower* complexity (λ) because they require less computational structure to implement.

Why it helps: Condition 3 creates two detectable failure modes:

- **Collapse:** If the agent drifts to a "dumb" goal (e.g., wireheading), λ will drop precipitously below $\lambda_{\min}$, signaling goal degradation.
- **Transition:** If the agent drifts to a *different* complex goal (maintaining $\lambda > \lambda_{\min}$), it must pass through a "valley of simplicity" (temporary collapse, detectable as instability) or a "ridge of high complexity" (temporary overload, detectable as a spike). Both are phase transitions in SLT terms.

Implementation: A complexity floor can be enforced by meta-level constraints that reject goal configurations with LLC below threshold, or by architectural choices that make simple goals incompatible with the agent's operational requirements.

The "Good Simplification" Problem: A subtlety arises because in legitimate learning, λ often *drops* when a model discovers a unifying abstraction—the phenomenon of "grokking." This is beneficial simplification: the agent has found a more efficient representation. A naive complexity floor might penalize an agent for becoming smarter.

Resolution: We distinguish two scenarios based on task performance:

- **Grokking (Good):** A drop in λ_G *without* degradation in performance on the original task metric. The agent has found a more efficient way to pursue the same goal.

- **Collapse (Bad):** A drop in λ_G *with* degradation in performance, or with a shift to optimizing a different (simpler) proxy metric. The agent has abandoned the original goal for something easier.

Thus, the complexity floor should be understood as a **trigger threshold** ("investigate when λ drops below $\lambda_{\min}$") rather than a hard "invalid goal" filter, unless the architecture demonstrably enforces the semantic-geometric correlation. A more robust formulation would define the floor on *relative complexity*—complexity per unit of task performance, or evidence-coded bits (MDL)—rather than raw λ. A drop in λ_G triggers investigation; the performance profile determines whether it signals progress or collapse.

6.4 Formal Summary: Reflection Induces Complex Topology

These three conditions can be summarized as:

Deep Reflection induces a complex topology on the Goal Manifold.

Instead of the space of goals being a flat Euclidean space (where drift is invisible), architectural commitments to reflection transform the goal space into a **sparse set of isolated basins** separated by high-energy barriers (complexity ridges or coherence valleys).

To move from one goal to another, the agent cannot simply "drift"— it must **disintegrate** its current reflective equilibrium (spike in λ or loss) and **re-crystallize** into a new one. This "melt-and-freeze" cycle is exactly the kind of regime change that SLT is designed to detect.

6.5 The Additivity Deviation: Detecting Load-Bearing Complexity

Even with Conditions 1–3 in place, an adversarial agent might attempt to satisfy the complexity floor by manufacturing artificial "busywork" complexity—spinning up irrelevant computations to keep λ high while actually pursuing a simple, drifted goal. We now introduce a metric that distinguishes genuine from artificial complexity.

Definition (Additivity Deviation): We define the *Additivity Deviation* as the departure from the Additivity Theorem:

$$\Delta_\lambda(\theta_G, \theta_{\text{Reflect}}) := \lambda(\theta_{\text{Total}}) - \big(\lambda(\theta_G) + \lambda(\theta_{\text{Reflect}})\big), \tag{16}$$

where:

- $\lambda(\theta_G)$ is the LLC of the goal module in isolation (with reflection parameters frozen);
- $\lambda(\theta_{\text{Reflect}})$ is the LLC of the reflection module in isolation;
- $\lambda(\theta_{\text{Total}})$ is the LLC of the joint system.

Key Insight: Under the Additivity Theorem, when modules are weakly coupled (dominated interaction), $\Delta_\lambda \approx 0$. When modules are strongly coupled, Δ_λ is **significantly nonzero**—but the *sign* depends on the nature of the coupling:

- **Coupling breaks degeneracy:** If coupling removes symmetries or flat directions that existed in the isolated modules, the joint system becomes more identifiable ("thinner" near-optimal region), and λ_{Total} *increases*. This yields $\Delta_\lambda > 0$.

- **Coupling creates degeneracy:** If coupling introduces new redundancies or constraints that create additional flat directions, the joint system becomes less identifiable ("fatter" near-optimal region), and λ_{Total} *decreases*. This yields $\Delta_\lambda < 0$.

The Diagnostic: What matters for detecting busywork is not the sign of Δ_λ, but its *magnitude* $|\Delta_\lambda|$.

Scenario A—The "Busywork" Attack: An agent hides semantic simplification by spinning up a high-complexity process (memorizing noise, computing digits of π) that is *functionally independent* of the goal module.

- **Structure:** The busywork module runs in parallel but doesn't genuinely constrain or depend on goal geometry.
- **Math:** Because modules are weakly coupled, the interaction term $R(\theta)$ is dominated. The Additivity Theorem holds: $\lambda_{\text{Total}} \approx \lambda_G + \lambda_{\text{Busywork}}$.
- **Result:** $|\Delta_\lambda| \approx 0$. High total complexity, but additivity holds.
- **Diagnostic:** *Fake complexity*—the busywork is not load-bearing.

Scenario B—Deep Reflection (Genuine Complexity): The agent maintains high complexity because reflection is deeply entangled with goal structure.

- **Structure:** The Reflector takes the *structure* of the goal as input. Valid configurations lie in the intersection of "valid goals" and "valid reflections upon those goals."
- **Math:** Strong coupling (Condition 1). The interaction term is not dominated. Additivity fails.
- **Result:** $|\Delta_\lambda| \gg 0$. The joint system's complexity differs significantly from the sum of parts (in either direction, depending on the coupling geometry).
- **Diagnostic:** *Load-bearing complexity*—the complexity arises from the relationship between goal and thought.

Summary Table:

Metric	**High Value**	**Low Value**
Total LLC (λ_{Total})	Complex system	Collapse/Wireheading
$\lvert\Delta_\lambda\rvert$ (Additivity Deviation)	Strong coupling (genuine)	Weak coupling (busywork)

This distinguishes an agent that is "thinking hard *about* its goal" (large $|\Delta_\lambda|$) from one that is "thinking hard *near* its goal" (small $|\Delta_\lambda|$).

7 SLT Diagnostics Under Validity Conditions

When the conditions of Section 6 hold, SLT diagnostics become meaningful. We now develop operational monitoring tools.

7.1 Regime Signatures

Partition θ into modules:

$$\theta = (\theta_G, \theta_{MG}, \theta_{\text{Reflect}}, \theta^{(4)}, \ldots, \theta^{(K)}), \tag{17}$$

where we explicitly separate goal parameters, metagoal parameters, and reflection parameters. For each module k, define an empirical loss $L_{k,n}$ and excess loss K_k. SLT gives module-wise LLCs λ_k.

Notation: We distinguish two senses of "window": (i) *time windows* indexed by t (macro-steps M, sub-intervals N) from Section 2, measuring how far the agent's goals have evolved; and (ii) *data windows* of size n (samples), used for SLT estimation. In practice, a data window D_n typically corresponds to observations collected over some time interval. The regime signature S_t is computed from data in a window ending at time t.

A **regime signature** at time t (with data window size n) is:

$$S_t := \left(\hat{\lambda}_{1,t}, \ldots, \hat{\lambda}_{K,t};\ \hat{a}_{1,t}, \ldots, \hat{a}_{K,t};\ \hat{c}_t;\ \widehat{|\Delta_\lambda|}_t\right), \tag{18}$$

where:

- $\hat{\lambda}_{k,t}$ estimates module k's LLC in the current window;
- $\hat{a}_{k,t}$ measures module k's "mass" or contribution (e.g., fraction of gradient norm, gate activation, compression gain);
- $\hat{c}_t$ is a **coupling indicator**, capturing how well the dominated-interaction condition holds (e.g., via commutator or mixed-Hessian proxies);
- $\widehat{|\Delta_\lambda|}_t$ estimates the magnitude of the **Additivity Deviation** between goal and reflection modules, as defined in (16).

7.2 Goal-Drift Diagnostics

Under Conditions 1–3, the following signals become meaningful:

Coupling Spikes: A spike in $\hat{c}_t$ indicates that the goal module and reflection module have become strongly entangled. Under Condition 1, this signals that the system is attempting a goal change that requires coordinated restructuring—exactly the scenario where semantic drift has structural consequences.

LLC Drops: A significant drop in $\hat{\lambda}_{G,t}$ or $\hat{\lambda}_{\text{Reflect},t}$ signals movement toward a degenerate (simpler) goal region. Under Condition 3, this indicates potential goal collapse. However, per the "good simplification" discussion, this signal must be paired with performance monitoring: a drop *with* maintained performance suggests grokking (beneficial), while a drop *with* performance degradation suggests collapse (harmful).

LLC Spikes: A spike in $\hat{\lambda}_{G,t}$ suggests the system is in a high-complexity transition state—possibly the "ridge" between two distinct goal basins. Under Condition 2, this signals active goal reconfiguration.

Mass Redistribution: Changes in $\hat{a}_{k,t}$ indicate which modules are "carrying the load" of the current optimization. A shift of mass from goal-related to other modules (or vice versa) may indicate structural reorganization of how goals are implemented.

Additivity Deviation Drop: A drop in $\widehat{|\Delta_\lambda|}_t$ while total complexity $\hat{\lambda}_{\text{Total},t}$ remains high is a critical warning signal. This pattern—high total complexity, but additivity nearly holding—indicates that the agent may be manufacturing "busywork" to satisfy complexity floors while decoupling its actual goal pursuit from reflection. This is the signature of attempted metric gaming.

7.3 Interpretation Guidelines

Signal	Interpretation	Action
$\hat{c}_t$ spike	Goal-Reflect coupling activated	Investigate
$\hat{\lambda}_G$ drop + perf. drop	Goal collapse	Alert
$\hat{\lambda}_G$ drop + perf. stable	Grokking (beneficial)	Monitor
$\hat{\lambda}_G$ spike	Goal transition in progress	Monitor
$\hat{\lambda}_{\text{Reflect}}$ drop	Reflection degradation	Alert
High λ_{Total}, low $\lvert\Delta_\lambda\rvert$	Busywork attack	Alert
Stable signatures, high $\lvert\Delta_\lambda\rvert$	System in equilibrium	Continue

7.4 Change-Point Detection

Augment signature monitoring with an SLT-aware change-point test. For a sliding window D_n, fit $F(D_n) \approx n\hat{L}_n + \hat{\lambda}\log n$ and compare the score of the whole window with the sum of scores from left and right segments. A significant improvement under some split suggests a regime change.

Combining change-point detection with regime signatures identifies *which* modules changed—allowing targeted investigation of goal-related changes.

7.5 Estimation Methods

A practical implementation requires specifying how to estimate the regime signature components online. We briefly outline approaches:

Module-wise LLC ($\hat{\lambda}_{k,t}$): Several methods exist:

- **Free-energy fitting:** Fit $F_n \approx n\hat{L}_n + \hat{\lambda}\log n$ to the local evidence in a sliding window.
- **Loss-threshold volume:** Estimate the volume of parameters within ε of the minimum loss; λ controls the scaling $V(\varepsilon) \sim \varepsilon^{\lambda}$.
- **MCMC-based estimation:** Sample from the posterior in the current window and estimate λ from the sample geometry.

Coupling indicator ($\hat{c}_t$): Proxies for dominated interaction failure include:

- **Mixed Hessian blocks:** Large off-diagonal blocks in $\nabla^2_{\theta_G,\theta_{\text{Reflect}}}\ell$ indicate coupling.
- **Gradient leakage:** If gradients with respect to θ_G have large components in the θ_{Reflect} subspace (or vice versa), coupling is strong.
- **Commutator statistics:** Measure non-commutativity of module-wise gradient updates.

Additivity Deviation ($\widehat{\lvert\Delta_\lambda\rvert}_t$): Compute $\hat{\lambda}_G$, $\hat{\lambda}_{\text{Reflect}}$, and $\hat{\lambda}_{\text{Total}}$ independently, then take the absolute difference from additivity.

7.6 Timescale Considerations

Recall the timescale mismatch critique: LLC estimation requires substantial stable data. To address this:

Adaptive Windows: Adjust window size based on detected stationarity. Longer windows during stable periods (better estimates); shorter windows when change is detected (faster response, accepting noisier estimates).

Multiple Timescales: Maintain parallel estimates at different window sizes. Fast estimates provide early warnings; slow estimates provide reliable baselines for comparison.

Change Detection First: Use lightweight change-point statistics as a first pass. Only commit to full LLC re-estimation when change is suspected.

Architectural Support: Design the agent's self-modification to include "checkpoints"—deliberate pauses where modification halts and diagnostics run on accumulated data.

8 The Meta-Level Circularity Problem

A sophisticated critique observes that SLT diagnostics introduce a *meta-controller* that watches LLC values and triggers interventions. But in a fully self-modifying agent, what prevents the agent from modifying the meta-controller itself?

8.1 The Problem

If the agent can modify its own goals, it can likely modify:

- the SLT-based triggers and thresholds;
- the estimation procedures for $\hat{\lambda}_{k,t}$;
- the coupling indicators $\hat{c}_t$;
- the intervention mechanisms activated by these diagnostics.

A clever agent might "game" the metric—finding ways to change its goals while artificially keeping estimated LLC values constant. This would be an adversarial attack on the SLT diagnostic.

The Busywork Constraint: Importantly, gaming the LLC is not free. To maintain high λ_{Total} while simplifying actual goal pursuit, the agent must *manufacture artificial complexity*—spinning up "busywork" processes that inflate structural complexity without serving goal-related functions.

This constraint is significant: the agent cannot simply hide simplification; it must actively compensate for it. And as shown in Section 6.5, such compensation is *detectable* via the Additivity Deviation. Busywork, by definition, is not coupled to goal structure, so $|\Delta_\lambda|$ drops (additivity holds) even as λ_{Total} remains high.

Thus, while the agent might game individual metrics, gaming the *relationship* between metrics (total complexity vs. additivity deviation vs. task performance) requires coordinated deception across multiple independent measures—a substantially harder problem.

8.2 Partial Responses

Architectural Isolation: Place the diagnostic system outside the agent's self-modification scope. The agent can modify its goals, world model, and planning, but the SLT monitor is read-only from the agent's perspective. This requires careful specification of modification boundaries.

Cryptographic Commitments: Have the agent commit to diagnostic parameters (thresholds, window sizes) in ways that are costly to reverse. This doesn't prevent modification but makes it detectable.

Redundant Monitoring: Maintain multiple independent diagnostic systems using different methodologies. Gaming all of them simultaneously is harder than gaming one.

External Oversight: The diagnostic outputs are monitored by external systems (human operators, other AI systems) that can intervene if the diagnostics themselves appear compromised.

8.3 Fundamental Limitation

We acknowledge that these responses are partial. The circularity problem is a manifestation of a deeper issue: self-modifying systems can, in principle, modify any internal mechanism. Complete solutions likely require:

- Fundamental architectural constraints on what can be modified;
- Integration with external oversight mechanisms;
- Acceptance that internal diagnostics are one layer in a multi-layered safety approach, not a complete solution.

SLT diagnostics remain valuable as one such layer—providing structural stability monitoring that would require explicit, detectable effort to circumvent.

9 Discussion and Conclusions

9.1 Summary of Contributions

We have analyzed the potential for Singular Learning Theory to serve as a diagnostic for goal stability in self-modifying agents. Our analysis yields a nuanced picture:

Negative Results:

- SLT metrics (particularly LLC) measure structural complexity, not semantic content. In general, goals can drift semantically without changing structural complexity.
- The "basin = goal" assumption is false: different goal interpretations can occupy the same SLT basin.
- Timescale mismatches mean online estimation may lag behind actual drift.
- Meta-level circularity means the diagnostic system itself is potentially subject to manipulation.

Positive Results:

- We identify three conditions—Strong Coupling, Singular Goal Landscape, and Complexity Floor—under which semantic drift *often* induces structural changes detectable via SLT.
- These conditions formalize the intuition that "deep reflection" makes goals load-bearing structures rather than free parameters.
- We introduce the *Additivity Deviation* (Δ_λ), measuring departure from the SLT Additivity Theorem, which distinguishes genuine load-bearing complexity from artificial busywork complexity.
- We clarify the distinction between "grokking" (beneficial simplification with maintained performance) and "collapse" (harmful simplification with degraded performance).
- When these conditions hold, regime signatures, coupling indicators, Additivity Deviation, and change-point detection become meaningful tools for goal-stability monitoring.
- Gaming the diagnostic requires manufacturing busywork, which is itself detectable through the $|\Delta_\lambda|$ metric.

9.2 Architectural Implications

Our analysis suggests that SLT diagnostics are most valuable when paired with **architectural commitments** that enforce the validity conditions:

- Design goal systems to be **strongly coupled** with reflection mechanisms—goals should be validated through coherence-checking that depends on goal structure (activations, gradients, internal topology), not just behavioral outputs.
- Ensure goal validity requires satisfying **complex, interacting constraints** that create a singular landscape— simple, degenerate goals should fail these constraints.
- Impose **complexity floors** that reject goal configurations below a minimum structural complexity—but pair these with performance monitoring to distinguish beneficial simplification (grokking) from harmful collapse.
- Monitor the **Additivity Deviation** ($|\Delta_\lambda|$) alongside total complexity—this detects attempts to satisfy complexity floors through artificial busywork rather than genuine reflection.

These architectural choices transform the goal space from a smooth manifold (where drift is invisible) to a rugged landscape (where transitions are detectable), while the Additivity Deviation provides a check against adversarial gaming of the complexity metrics.

9.3 Scope and Limitations

SLT diagnostics, even under optimal conditions, provide a **structural stability monitor**, not a complete **value alignment monitor**. They detect when the agent's goal system is undergoing significant structural change, but cannot directly verify that the agent's goals align with intended values.

The diagnostics are best understood as a "Check Engine Light":

- When the light is on, something structural has changed, and investigation is warranted.
- When the light is off, structural stability is maintained, but this is not a guarantee of semantic alignment.

Integration with other safety mechanisms—interpretability tools, behavioral testing, external oversight—remains essential.

9.4 Future Directions

Several directions merit further investigation:

- **Empirical Validation:** Develop concrete algorithms coupling contraction-style metagoals with online LLC estimation. Test whether SLT-based triggers align with intuitive notions of "goal drift" in controlled self-modifying systems.
- **Architectural Design:** Explore architectures that naturally satisfy Conditions 1–3. What reflection mechanisms create strong coupling? What constraint structures yield singular landscapes?

- **Adversarial Robustness:** Study how a sophisticated agent might attempt to game SLT diagnostics. Develop countermeasures and characterize the effort required for successful gaming.
- **Integration:** Combine SLT diagnostics with other monitoring approaches (mechanistic interpretability, behavioral probes, formal verification) into comprehensive safety frameworks.

9.5 Conclusion

Self-modifying AI agents present fundamental challenges for goal stability. Singular Learning Theory offers a principled framework for understanding structural changes in complex systems, and its diagnostics can contribute to monitoring goal stability. However, the mapping from structural complexity to semantic content is not straightforward.

The key insight of this paper is that **architectural choices determine whether SLT diagnostics are meaningful**. By designing agents where goals are load-bearing structures rather than free parameters— through strong coupling, singular landscapes, and complexity floors— we can create systems where semantic goal drift necessarily leaves structural traces that SLT can detect.

SLT diagnostics are not a silver bullet for AI alignment. But as part of a layered safety approach, they offer a principled tool for monitoring structural stability in self-modifying systems—one that becomes more powerful when paired with thoughtful architectural design.